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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (Electrical Engg./Mechanical Engg.) (Sem.–7,8)

MECHATRONICS SYSTEMS

Subject Code : BTME-610-18

M.Code : 90478

Date of Examination : 14-07-22

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

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SECTION-A

1. Write briefly :

a. What is mechatronic approach?

b. What is photovoltaic transducer?

c. Explain *"Latching"* with ladder diagram.

d. Differentiate between mechatronics and traditional system design.

e. List down the input and output modules interface.

f. Write a short note on micro-sensors.

g. Differentiate between microprocessor and microcontroller.

h. How PLC is different from minicomputer.

i. What are magneto actuators?

j. Define reproducibility in sensors.

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SECTION-B

2. Explain the basics of ladder programming used in PLC's.

3. Explain the working principle of stepper motor in half step mode.

4. Differentiate between pneumatic and hydraulic actuators.

5. Explain the static characteristics of transducers.

6. Explain the factors for selecting the micro-controller.

SECTION-C

7. Explain about the design of mechatronic systems with any example.

8. Explain the position and velocity control loops for continuous path control.

9. Discuss the role of sensors in car engine management system. Explain with a block diagram.

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NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.